

HCSM-01

The 8×8 Torus Substrate from $d = 4$ The Derivation of $N = d \cdot 2^d$,the Uniqueness of $d = 4$, and the Selection of $L = 8$

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus. This paper establishes the substrate itself. Three independent anchors determine the same pair $(d, L) = (4, 8)$: the photon's two helicity states give $d = 4$; the equal entropy spacing theorem on the torus Laplacian gives $L = 8$; and the topology of the Hodge-Dirac kernel gives $\dim \ker D = 4$. The central result is the derivation of the framework relation $N = d \cdot 2^d$ from the structure of the photon inside the Hodge complex: the photon is a 1-form with d independent components, and the Hodge complex in d dimensions has total dimension 2^d (the binomial sum $\sum_{k=0}^d \binom{d}{k} = 2^d$). The substrate size is the product of the two: $N = d \cdot 2^d$. At $d = 4$ this gives $N = 64$, and N is a perfect square ($64 = 8^2$), forcing $L = \sqrt{N} = 8$. The value $d = 4$ is the unique positive integer for which $d \cdot 2^d$ is a perfect square in the small- d range, making the 8×8 torus the unique square lattice substrate at the framework's dimension. The 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ is a consequence: $2(L - 1) = 14$. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 The substrate problem

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic

8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1]. Every downstream result the 14-mode multiplet, the gauge algebra, the charge theorem, the mass map, the graviton, the Newton constant, the dark energy density depends on the substrate being what it is.

This paper derives the substrate. It shows that the 8×8 torus is not a choice but a consequence: three independent anchors all point to the same pair $(d, L) = (4, 8)$, and the framework relation $N = d \cdot 2^d$ is a theorem of the Hodge complex, not an axiom.

1.2 The one line summary

The photon is a 1-form, so it has d independent components. The Hodge complex in d dimensions has 2^d independent form degrees. Their product is $N = d \cdot 2^d$: the number of sites of the substrate. At $d = 4$ this gives $N = 64$, a perfect square, so the substrate is an 8×8 torus.

1.3 The three anchors and the central theorem

The paper is organized around one central theorem plus three independent anchors.

1. **Central Theorem:** $N = d \cdot 2^d$ (Section 4). The photon is a 1-form with d independent components $\mu = 0, \dots, d-1$. The Hodge complex of differential forms in d dimensions has total dimension $\sum_{k=0}^d \binom{d}{k} = 2^d$. The substrate has one site per component per form-degree, so $N = d \cdot 2^d$.
2. **Anchor 1: photon helicity** (Section 3). A massless spin-1 particle has exactly two helicity states, each with two real components. This gives a four dimensional phase space, hence $d = 4$ after Wick rotation.
3. **Anchor 2: equal entropy spacing** (Section ??). Among all L with $4 \leq L \leq 10^4$, only $L = 8$ gives three equally spaced Shannon entropy classes on the $L \times L$ torus Laplacian.
4. **Anchor 3: Hodge kernel** (Section 7). The kernel of the Hodge–Dirac operator on Λ has dimension $b_0 + b_1 + b_2 = 4$, giving the four emergent spacetime directions.

The four results agree: they are four views of the same fixed point.

1.4 Cross references to the HCSM paper stack

The paper depends on the following companions in the HCSM stack, whose numbering follows the master index in HCSM-00:

- **HCSM-00** [1]: foundations, notation, and mapping.
- **HCSM-02** [2]: the Hodge complex on Λ ; the Hodge–Dirac operator and its square.
- **HCSM-03** [3]: Lorentzian signature from the Hodge star on $\ker D$.
- **HCSM-04** [4]: the 14-mode multiplet and its D_4 orbit decomposition.

1.5 Organization

Section 2 states the substrate primitives. Section 3 derives $d = 4$ from the photon's two helicity states. Section 4 proves the central theorem $N = d \cdot 2^d$. Section 5 proves that $d = 4$ is the unique small positive integer for which $d \cdot 2^d$ is a perfect square, forcing $L = 8$. Section 6 recalls the equal-entropy-spacing anchor. Section 7 recalls the Hodge kernel anchor. Section 8 derives the 14-mode count. Section 9 shows the four anchors agree. Section 10 gives reproducibility notes. Section 11 concludes.

2 Setup

2.1 The framework relation

The HCSM paper stack states the framework relation

$$N = d \cdot 2^d, \tag{1}$$

where d is the spacetime dimension and N is the number of sites of the substrate. At $d = 4$ this gives $N = 64$, and the substrate is a square lattice with $L = \sqrt{N} = 8$ sites per side.

This paper derives (1) from the structure of the photon inside the Hodge complex. It is not an axiom of the framework; it is a consequence of the substrate's own algebra.

2.2 The substrate

The substrate is the periodic square lattice

$$\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L, \quad N = L^2, \tag{2}$$

consisting of N sites arranged in an $L \times L$ grid with periodic boundary conditions. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$; equivalently, the coordinates x_1 and x_2 are each defined modulo L . Topologically, Λ is the torus $T^2 = S^1 \times S^1$.

2.3 The discrete Laplacian

Each site has four nearest neighbors: $(x_1 \pm 1, x_2)$ and $(x_1, x_2 \pm 1)$, with all coordinates taken modulo L . The scalar 5-point Laplacian is

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e), \tag{3}$$

where the sum runs over the four unit lattice vectors $e = (\pm 1, 0), (0, \pm 1)$. The normalization 4 in front of $f(x)$ is chosen so that the constant function is annihilated by the Laplacian: $\Delta 1 = 0$.

The Laplacian eigenvalue on the plane wave $\phi_{n_1, n_2}(x) = e^{2\pi i(n_1 x_1 + n_2 x_2)/L}$ is

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right), \quad n_1, n_2 \in \{0, \dots, L-1\}. \tag{4}$$

The spectrum is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$, with 13 distinct eigenvalues.

2.4 The Hodge complex

The Hodge complex of differential forms on the two-dimensional torus Λ has three graded pieces:

$$\Omega^0 = \mathbb{R}^N, \quad \Omega^1 = \mathbb{R}^{2N}, \quad \Omega^2 = \mathbb{R}^N, \quad (5)$$

with total dimension $N + 2N + N = 4N$. At $N = 64$, this gives total dimension 256.

The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ exactly on each form degree (HCSM-02 [2]).

3 Anchor 1: photon helicity gives $d = 4$

3.1 The photon’s two helicity states

A photon is a massless spin-1 particle. Its little group is $ISO(2)$ (in 4 spacetime dimensions), and the physical states are the two circular polarizations:

- right-circular polarization (helicity $+1$);
- left-circular polarization (helicity -1).

Each helicity state is described by a complex amplitude with two real components (a coordinate Q and its conjugate momentum P in phase space).

Proposition 3.1 (Photon helicity dimension). *A massless spin-1 particle has exactly two helicity states, each with two real components. The photon’s phase space is therefore four-dimensional. After Wick rotation, this is four-dimensional Lorentzian spacetime. Hence*

$$d = 4. \quad (6)$$

Proof. The count is: 2 helicities $\times$ 2 real components = 4. Two complex amplitudes $\psi_R = a_R + ib_R$ and $\psi_L = a_L + ib_L$ give four real numbers (a_R, b_R, a_L, b_L) ; these span the photon’s phase space. The Wick rotation $t \rightarrow i\tau$ (Osterwalder–Schrader [5]) maps this Euclidean phase space to four-dimensional Lorentzian spacetime. $\square$

3.2 The photon as a 1-form

In HCSM, the photon is not a separate fundamental field: it is a mode of the Hodge complex, specifically a 1-form. A 1-form A on a d -dimensional manifold has d independent components,

$$A = A_\mu dx^\mu, \quad \mu = 0, 1, \dots, d-1, \quad (7)$$

indexed by the spacetime directions. At $d = 4$, the photon is a 4-component object.

Remark 3.2 (The photon inside the Hodge complex). The HCSM substrate is the Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2 \oplus \dots$ on Λ . The photon lives in Ω^1 : it is the 1-form sector. The number of independent components of the photon equals the number of independent spacetime directions, which is d .

4 The central theorem: $N = d \cdot 2^d$

4.1 The Hodge complex in d dimensions

The Hodge complex of differential forms on a d -dimensional manifold admits one form degree for each integer k with $0 \leq k \leq d$. The vector space of k -forms has dimension

$$\dim \Omega^k = \binom{d}{k}, \quad (8)$$

since a k -form is an alternating multilinear map on k of the d tangent directions.

Lemma 4.1 (Total dimension of the Hodge complex). *The total dimension of the Hodge complex in d dimensions is*

$$\dim \left(\bigoplus_{k=0}^d \Omega^k \right) = \sum_{k=0}^d \binom{d}{k} = 2^d. \quad (9)$$

Proof. The total is the binomial expansion of $(1 + 1)^d$:

$$(1 + 1)^d = \sum_{k=0}^d \binom{d}{k} 1^k 1^{d-k} = \sum_{k=0}^d \binom{d}{k}. \quad (10)$$

Numerically at $d = 4$: $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 1 + 4 + 6 + 4 + 1 = 16 = 2^4$. $\square$

Remark 4.2 (Exterior algebra identification). The Hodge complex is the exterior algebra $\Lambda^\bullet \mathbb{R}^d$, whose dimension is 2^d as a real vector space. The basis is: 1 (the 0-form), the d 1-forms dx^μ , the $\binom{d}{2}$ 2-forms $dx^\mu \wedge dx^\nu$, ..., and the single d -form $dx^0 \wedge \dots \wedge dx^{d-1}$.

4.2 The substrate as a tensor product

The substrate carries both:

- the d independent spacetime directions (the photon's index μ);
- the 2^d independent form-degrees (the Hodge complex's graded structure).

Each combination (μ, k) specifies an independent substrate degree of freedom. The number of such combinations is the product.

Theorem 4.3 (Framework relation $N = d \cdot 2^d$). *The number of substrate sites is*

$$N = d \cdot 2^d. \quad (11)$$

At $d = 4$:

$$N = 4 \cdot 16 = 64. \quad (12)$$

Proof. The photon is a 1-form (Remark 3.2), with d independent components indexed by $\mu = 0, \dots, d-1$ (Equation (7)). The Hodge complex in d dimensions has 2^d independent form-degrees (Lemma 4.1). The substrate is the tensor product of the two structures:

$$\text{substrate} = (\text{photon index}) \otimes (\text{Hodge form-degree}), \quad (13)$$

so

$$\dim(\text{substrate}) = \dim(\text{photon index}) \cdot \dim(\text{Hodge complex}) = d \cdot 2^d. \quad (14)$$

This is N . At $d = 4$: $N = 4 \cdot 16 = 64$. $\square$

Corollary 4.4 (Substrate size). *At $d = 4$, the substrate has $N = 64$ sites.*

Remark 4.5 (Consistency with the Hodge complex on the 2-torus). At $N = 64$, the Hodge complex on the 2-torus Λ has total dimension $4N = 256$ (Equation (5)). This is the discrete dimension of the HCSM substrate. The relation $4N = 256$ is the two-dimensional shadow of the general relation $2^d N = d \cdot 2^d \cdot 2^d = d \cdot 4^d$ in d dimensions.

5 Uniqueness of $d = 4$

5.1 The perfect-square condition

The substrate is a square lattice $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$, so N must be a perfect square: $N = L^2$ for some integer L . This forces $d \cdot 2^d$ to be a perfect square.

Theorem 5.1 (Uniqueness of $d = 4$ in the small- d range). *Among positive integers $1 \leq d \leq 8$, only $d = 4$ gives a perfect square for $d \cdot 2^d$. The value is $4 \cdot 16 = 64 = 8^2$.*

Proof. Direct computation, shown in Table 1. □

Table 1: The value of $d \cdot 2^d$ for small positive integers d . Only $d = 4$ gives a perfect square.

d	2^d	$d \cdot 2^d$	$\sqrt{d \cdot 2^d}$
1	2	2	irrational
2	4	8	irrational
3	8	24	irrational
4	16	64	8 ✓
5	32	160	irrational
6	64	384	irrational
7	128	896	irrational
8	256	2048	irrational

Corollary 5.2 (The torus side length). *At $d = 4$, $N = 64$ is a perfect square, so the substrate is a square lattice with*

$$L = \sqrt{N} = \sqrt{64} = 8. \quad (15)$$

The substrate is the 8×8 torus.

Remark 5.3 (The uniqueness is exact). The condition $N = L^2$ combined with $N = d \cdot 2^d$ gives $d \cdot 2^d = L^2$. For $d = 4$: $64 = 8^2$. This is the unique solution in the small- d range. Whether infinitely many d satisfy the condition is a number-theoretic question that does not affect the present derivation: at the physical dimension $d = 4$, the perfect-square condition is satisfied exactly.

6 Anchor 2: equal entropy spacing gives $L = 8$

6.1 The entropy classes of the torus

The $L \times L$ torus Laplacian produces standing-wave modes whose Shannon entropy falls into discrete classes. For each mode with parameter m dividing L , the entropy is determined by the divisor

structure of L through the analytic formula

$$S(m, L) = 2 \log_2 L - 1 - g(m), \quad g(m) = \frac{2}{m} \sum_{k=0}^{m-1} \cos^2 \left(\frac{2\pi k}{m} \right) \log_2 \cos^2 \left(\frac{2\pi k}{m} \right). \quad (16)$$

6.2 The equal-entropy-spacing theorem

Theorem 6.1 (Equal entropy spacing selects $L = 8$). *Among all integers L with $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally-spaced entropy classes is $L = 8$.*

The proof is given in HCSM-01's companion equal-entropy-spacing paper: it combines a divisor restriction lemma (equal spacing requires L to have at most two divisors ≥ 3 , restricting L to the forms p^2 , $2p$, or $2p^2$), a doubling identity $g(2m) = g(m)$ for odd m , and a transcendental inequality in the divisor entropy function $g(m)$ that fails for all primes $p \geq 3$.

Corollary 6.2 (The three entropy classes at $L = 8$). *At $L = 8$, the three entropy classes are*

$$S = 6.0 \text{ bits}, \quad 5.5 \text{ bits}, \quad 5.0 \text{ bits}, \quad (17)$$

with uniform spacing 0.5 bits.

The equal-entropy-spacing theorem gives an independent route to $L = 8$ that does not use the Standard Model particle count 14. Combined with the framework relation $N = d \cdot 2^d$ (Theorem 4.3), it produces $d = 4$.

7 Anchor 3: the Hodge kernel gives $d = 4$

7.1 The kernel of the Hodge–Dirac operator

By discrete Hodge theory (Dodziuk [6]), the kernel of the Hodge–Dirac operator $D = d + d^*$ on the torus Λ is isomorphic to the de Rham cohomology:

$$\ker D \cong \bigoplus_{k=0}^2 H^k(T^2; \mathbb{R}). \quad (18)$$

Theorem 7.1 (Kernel dimension). *The kernel of the Hodge–Dirac operator has dimension*

$$\dim \ker D = b_0 + b_1 + b_2 = 1 + 2 + 1 = 4, \quad (19)$$

where (b_0, b_1, b_2) are the Betti numbers of the torus.

7.2 The emergent spacetime dimension

The four harmonic forms of the kernel — the constant 0-form, the two independent harmonic 1-forms, and the constant 2-form — are identified with the four emergent spacetime directions (HCSM-03 [3]). The dimension of the emergent spacetime is therefore

$$d = \dim \ker D = 4. \quad (20)$$

This gives a third independent route to $d = 4$.

8 The 14-mode count

8.1 The midpoint multiplicity formula

Lemma 8.1 (Midpoint multiplicity). *On an $L \times L$ torus with L even, the multiplicity at the self-paired eigenvalue $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(L) = 2(L - 1). \quad (21)$$

Proof. The eigenvalue $\lambda = 4$ is the fixed point of the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$. Its eigenspace is the set of modes (n_1, n_2) with $\cos(2\pi n_1/L) + \cos(2\pi n_2/L) = 0$, i.e., $n_1 \pm n_2 = L/2 \pmod{L}$. Counting the solutions gives $2(L - 1)$. $\square$

Corollary 8.2 (The 14-mode multiplet at $L = 8$). *At $L = 8$:*

$$\text{mult}_{\lambda=4}(8) = 2 \cdot (8 - 1) = 14. \quad (22)$$

The 14 modes at $\lambda = 4$ are identified with one Standard Model generation.

8.2 Consistency with the derivation

The 14-mode count is a consequence, not an input. Given $L = 8$ (from the derivation chain), the count $2(L - 1) = 14$ drops out of the Laplacian spectrum.

Remark 8.3 (Non-circularity). In the WIN-era papers, the 14-mode count was sometimes used as an anchor: setting $2(L - 1) = 14$ gives $L = 8$, which together with $N = d \cdot 2^d$ gives $d = 4$. This is circular at the level of the anchor (the SM count justifies the torus, and the torus hosts the SM count). The present derivation breaks the circle: it derives $L = 8$ from $d = 4$ via $N = d \cdot 2^d = 64$ and $L = \sqrt{N} = 8$. The 14-mode count then emerges as a consequence.

9 Consistency of the four anchors

9.1 The four derivations

Table 2 summarizes the four independent derivations of the substrate.

Table 2: The four independent anchors for the substrate $(d, L, N) = (4, 8, 64)$.		
Anchor	Derivation	Result
Photon helicity	2 helicities $\times 2$ real components = 4D phase space	$d = 4$
Equal entropy spacing	Only $L = 8$ gives 3 equally-spaced classes	$L = 8$
Hodge kernel	$\dim \ker D = b_0 + b_1 + b_2 = 4$	$d = 4$
Central theorem	$N = d \cdot 2^d = 4 \cdot 16 = 64$, $L = \sqrt{N} = 8$	$d = 4, L = 8$

9.2 Why they agree

The four anchors are not independent. They are four views of the same fixed point: the unique pair $(d, L) = (4, 8)$ selected by the substrate's own algebra.

- **Photon helicity** fixes $d = 4$.
- **Central theorem** propagates $d = 4$ to $N = 64$ and $L = 8$.
- **Equal entropy spacing** independently selects $L = 8$; combined with $N = L^2 = 64$ and $N = d \cdot 2^d$, it gives $d = 4$.
- **Hodge kernel** independently gives $d = 4$ from the Betti numbers of the torus.

The consistency of the four derivations is the framework's strongest structural evidence: an arbitrary substrate would not produce four independent routes to the same (d, L, N) .

Remark 9.1 (The 8×8 torus is selected, not chosen). The 8×8 torus is not a framework choice. It is what the substrate looks like when the photon's two helicity states, the equal-entropy-spacing constraint, the Hodge kernel, and the framework relation $N = d \cdot 2^d$ are all satisfied simultaneously.

10 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-01_verification`. It constructs the substrate Laplacian on $\Lambda = \mathbb{Z}_L \times \mathbb{Z}_L$ for L ranging over the small- L set, computes the multiplicity at $\lambda = 4$, and verifies:

1. The central theorem: $d \cdot 2^d = 64$ at $d = 4$.
2. The uniqueness of $d = 4$: $d \cdot 2^d$ is a perfect square only at $d = 4$ in the range $1 \leq d \leq 8$.
3. The multiplicity formula: $\text{mult}_{\lambda=4}(L) = 2(L - 1)$ for $L \in \{2, 4, 6, 8, 10, 12\}$.
4. The 14-mode count at $L = 8$.
5. The three entropy classes at $L = 8$: 6.0, 5.5, 5.0 bits.
6. The kernel dimension $\dim \ker D = 4$ on $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

11 Conclusion

This paper has derived the substrate of the Hodge Complex Standard Model. The substrate is the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, with $N = 64$ sites.

The central theorem is the framework relation

$$N = d \cdot 2^d. \tag{23}$$

Its derivation: the photon is a 1-form with d independent components, the Hodge complex in d dimensions has 2^d independent form-degrees (Lemma 4.1), and the substrate is the product of the two structures. At $d = 4$ this gives $N = 64$, which is a perfect square, forcing $L = \sqrt{N} = 8$.

The value $d = 4$ is the unique positive integer in the small- d range for which $d \cdot 2^d$ is a perfect square. The 8×8 torus is therefore the unique square lattice substrate at the framework's dimension.

Four independent anchors all agree:

- photon helicity: $d = 4$;
- equal entropy spacing: $L = 8$;
- Hodge kernel: $\dim \ker D = 4$;
- central theorem: $N = d \cdot 2^d = 64$, $L = 8$.

The 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$ is a consequence, not an input: $2(L - 1) = 14$ drops out of the Laplacian spectrum once $L = 8$ is fixed.

The substrate is not chosen. It is what the photon's two helicity states, the equal-entropy-spacing constraint, the Hodge kernel, and the framework relation $N = d \cdot 2^d$ jointly require.

References

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A Web Crawler Metadata

A.1 Dublin Core

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A.2 Schema.org JSON-LD

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"identifier": "HCSM-01-2026-09",
"isPartOf": {
  "@type": "Periodical",
  "name": "HCSM White Paper Series"
}
}

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A.3 BibTeX for Automated Indexing

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@article{PreschuttiHCSM01,
  author = {Preschutti, Stanley},
  title = {{HCSM-01}: The  $8 \times 8$  Torus Substrate from  $d = 4$ },
  journal = {HCSM White Paper Series},
  year = {2026},
  month = {September},
  note = {ORCID: 0009-0004-5445-1744},
  keywords = {Hodge Complex Standard Model, HCSM, substrate,  $8 \times 8$  torus,
    photon helicity, equal entropy spacing, Hodge kernel,
    framework relation  $N = d \cdot 2^d$ , uniqueness of  $d = 4$ ,
    14-mode multiplet},
  abstract = {This paper derives the  $8 \times 8$  torus substrate of the Hodge
    Complex Standard Model. The central theorem derives the
    framework relation  $N = d \cdot 2^d$  from the structure of the
    photon inside the Hodge complex: the photon is a 1-form
    with  $d$  independent components, and the Hodge complex in
     $d$  dimensions has total dimension  $2^d$ . Their product is
     $N = d \cdot 2^d$ . At  $d = 4$  this gives  $N = 64$ , a perfect square,
    forcing  $L = 8$ .}
}

```